

Delhi Community Radar

An AI-powered system that turns scattered civic complaints into prioritised action for Delhi.

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1 Abstract

Delhi Community Radar is a web prototype that lets residents report neighbourhood problems (garbage, waterlogging, broken streetlights, air pollution, or an elderly person who needs help) by **voice, photo or text** in **Hindi, English or Hinglish**. An AI model reads each report, finds the issue type and urgency, merges duplicate reports, drafts a formal complaint for the right Delhi authority, and suggests nearby volunteers. Reports appear on a live map and a priority list so the most urgent problems, and the most vulnerable people, are seen first. A password-protected admin console lets a moderator delete offensive content. The prototype runs on clearly labelled sample data.

2 Problem statement

Delhi's civic services are shared between many bodies (MCD, NDMC, the Delhi Jal Board, PWD, DPCC and others). A resident often has to work out who is responsible before complaining, and the same problem may be reported many times while urgent cases get lost in a long list. Reporting in English by typing a formal complaint is also hard for many people. Official channels such as the MCD 311 app and a unified 311 helpline exist [2][10]; this project is meant to **complement** them.

PROBLEM STATEMENT

How can residents of Delhi report civic and community problems easily, in their own language, so that duplicates are merged, urgent and vulnerable cases are ranked first, and every problem reaches the right authority or volunteer with a ready-to-send complaint?

3 Is this a genuine problem? Evidence from public sources

Area	What the source reports
Many agencies	Delhi launched a unified 311 helpline in May 2025 for NDMC, MCD, DDA, PWD, Flood Control and DJB. The minister said: "Citizens won't have to worry about which agency is responsible." [2] An analysis lists seven bodies with overlapping duties and says accountability is unclear [3] (2010).

Scale of complaints	330,099 civic complaints were recorded in 2019 (MCD 129,965; DJB 200,134) [1]. In 2024 the Green Delhi app received 84,765 pollution complaints; MCD had 8,599 still pending when reported [4].
No tracking (2020)	Praja Foundation found that "MCD has no mechanism for citizens to track their complaints" [1]. (Official apps have been launched since.)
Garbage	Delhi produces about 11,000 to 12,000 tonnes of fresh waste daily; Okhla and Bhalswa landfills are targeted for clearance by December 2026 and Ghazipur by December 2027. A resident says: "During monsoon, this black water reaches the residences." [5]
Waterlogging	On 20 Aug 2024 a school bus was trapped in the flooded Minto Bridge underpass and three children were rescued [6]. The underpass floods repeatedly and a person drowned there in July 2020 [7].
Air pollution	A <i>Lancet Planetary Health</i> study found Delhi had the most PM2.5-linked deaths of ten Indian cities: about 12,000 a year [8].
Older people	In a 20-city survey of 5,169 older people, 7% reported abuse, 33% had no income in the past year and only 9% knew the law that protects them [9].

Notes: the older-people survey is national, not Delhi-only, and no reliable Delhi-only figure was found. Some sources are older (2010, 2020) and are used for the structure of the problem. Complaint figures are as reported by the bodies or news outlets quoted.

4 The solution

How a report travels:

1. **Report** by typing, speaking (browser speech recognition), adding a photo, and setting the location by GPS or a tap on the map.
2. **Understand:** the AI returns the category (8 types), severity 1 to 5, locality, whether vulnerable people are affected, a summary and a suggested next step.
3. **Group:** reports of the same category within 700 m merge into one issue.
4. **Prioritise:** each issue gets a score from 0 to 100 (below).
5. **Route and draft:** the issue is matched to the responsible body (for example MCD, DJB, PWD, DPCC or a local RWA/NGO) and a formal complaint is drafted for the citizen to review and send.
6. **Match:** the two nearest volunteers with the right skill are suggested, and everything shows on the live map and priority list.

priority = (severity × 12) + (8 × merged duplicates) + (15 if vulnerable people are affected), maximum 100

Bands: **80+** critical, **60 to 79** high, **40 to 59** medium, **below 40** low. Example: an elderly person living alone in Karol Bagh, severity 3, merged with 1 similar report, vulnerable flagged: (3 × 12) + 8 + 15 = **59**.

Other features: a **Smog Watch** advisory for elderly people, children and outdoor workers (air-quality values are simulated in the prototype); a **flag button** so citizens can report offensive content; and an **admin console** for moderators. The prototype opens with 16 sample reports in 17 Delhi localities and 9 sample volunteers; none are real people or real complaints.

5 Technology and safety

Part	What is used
Front end	HTML, CSS and JavaScript; Leaflet + OpenStreetMap map; browser Web Speech API for voice.
Back end	Python with FastAPI. Data is kept in memory in the prototype.
AI	Claude (Anthropic) API for text and photo understanding and complaint drafting, used when an API key is set. Otherwise a keyword-based fallback runs, so the demo never breaks.

- **AI output is checked:** the category and locality must come from allowed lists and severity is limited to 1 to 5. The complaint is only a draft for the citizen to review; the system does not contact any authority itself.
- **Moderation:** reports with abusive words (English and Hinglish list), or flagged by the AI as abuse or spam, are held back and not shown publicly. Citizens can also flag a report, and it is hidden after flags from three different network addresses.
- **Admin console (/admin):** locked by a password; five wrong tries block login for five minutes. The administrator can **delete**, **hide** or **approve** a report, and every action is logged. All user text is escaped before display to block script injection.

6 Limitations (honest notes)

WHAT THE PROTOTYPE DOES NOT DO YET

- **It does not send complaints** to any authority; it drafts them for the resident to send. There is no integration with MCD, DJB or others.
 - **It does not contact volunteers;** it only suggests the nearest matches.
 - **It does not track resolution;** there is no status timeline yet.
- Reports, volunteers and air-quality values are **sample data**; live air quality would need an official feed such as CPCB.
 - If no location is given or recognised, the prototype assigns a placeholder locality. A real system must require a location.
 - AI can misread reports or photos. The full AI mode needs an API key and was not tested at scale. The fallback covers only 8 categories and 17 localities.
 - The routing of each category to an authority is the author's assumption and must be verified before real use.
 - Location and photos are sensitive; a real deployment would need consent, limited storage time and compliance with India's Digital Personal Data Protection Act, 2023.
 - This is **not an emergency service**; in an emergency, call 112.

7 Future scope

- **Close the loop (next):** tracking ID and status timeline; volunteer Accept button; proof of resolution with before and after photos compared by AI; escalation if an issue stays open too long.

- WhatsApp reporting, live CPCB or OpenAQ air-quality data, a real database with accounts and consent.
- Official portal integration where permitted, and ward-level analytics for officials and NGOs.
- A small pilot with one Resident Welfare Association and an NGO, measuring time to reach the right office and how easy reporting felt.

8 Conclusion

Public sources show that Delhi's civic and community problems are real, recurring and spread across many agencies, and that the government has already moved to simplify complaints. Delhi Community Radar is a small working step in that direction: it makes reporting easier through voice, photo and local languages, turns many scattered reports into one ranked picture, gives extra weight to vulnerable people, and is open about its limits. The next step is to close the loop with tracking and resolution, and to test it with real residents.

9 Prototype screenshots

STEP 1 / TELL US WHAT'S WRONG HI · EN · HINGLISH

Type here in Hindi, English or Hinglish. Example: Minto Bridge pe paani bhar gaya hai

UNDERSTANDING LOCATING ROUTING MATCHING HELP

PRIORITY 59

ELDERLY & VULNERABLE HELP · KAROL BAGH

Thanks, your report is in.

ROUTE TO
Local RWA / NGO → Delhi Social Welfare Dept.

HELP NEARBY
Aarti Sharma (0.1 km), Seva Delhi NGO (3.8 km)

NEXT STEP
Report logged and prioritised. Residents can document the issue with photos; suggested volunteers are listed.

DUPLICATES
Merged with 1 similar report(s)

VIEW DRAFTED COMPLAINT

To,
The Officer-in-Charge, Local RWA / NGO → Delhi Social Welfare Dept.

Subject: Urgent civic issue - Elderly & Vulnerable Help at Karol Bagh

Residents of Karol Bagh have reported the following: "Karol Bagh mein ek bujurg akele hain, tabiyat kharab hai, medicine chahiye". This is affecting daily life and public safety in the area. We request inspection and resolution within 48 hours and a status update to the complainant.

Regards,
Delhi Community Radar (on behalf of residents)

TRY AN EXAMPLE · VOICE LANGUAGE

SEE WHAT THE AI DID

Figure 1. Reporting: three input methods, progress bar, and the result with the suggested authority, volunteers and drafted complaint.

MAP	PRIORITY LIST (10)	Most urgent first · flags offensive reports
	71 Garbage & Waste	flag
	GHAZIPUR Ghazipur landfill ke paas kachra ka pahad, bahut badbu aur dhuhan hai, bachche بیمار ho rah...	
	2 reports merged · vulnerable people at risk	
	59 Elderly & Vulnerable Help	flag
	KAROL BAGH Karol Bagh mein ek bujurg akele rehte hain, unki tabiyat kharab hai, medicine chahiye	
	2 reports merged · vulnerable people at risk	
	51 Air Pollution	flag
	BHALSWA Bhalswa landfill fire smoke covering the colony, children coughing	
	vulnerable people at risk	
	51 Elderly & Vulnerable Help	flag
	SEELAMPUR Seelampur mein homeless elderly people sleeping outside in the cold smog, need blankets	
	vulnerable people at risk	
	44 Waterlogging & Drainage	flag
	CONNAUGHT PLACE Minto Bridge underpass mein paani bhar gaya hai, gaadiyan phans gayi hain. Urgent!	
	2 reports merged	
	36 Waterlogging & Drainage	flag
	DWARKA Dwarka mor drain overflow, sewer water on the road	
	36 Road & Potholes	flag
	OKHLA Okhla pothole on main road, two-wheeler accident today	
	36 Streetlight & Electricity	flag
	JANAKPURI Janakpuri mein transformer sparking near the park, danger	
	24 Streetlight & Electricity	flag
	DWARKA Streetlight kharab hai Dwarka sector 12, raat ko andhera, women safety issue	
	24 Water Supply	flag
	ROHINI Rohini sector 7 mein DJB ka paani gandha aa raha hai, 3 din se	

Figure 2. Priority list: most urgent first, with merged-report counts and a flag button.

ADMIN CONSOLE · RESTRICTED PUBLIC SITE

MODERATION

Report review

TOTAL REPORTS	NEEDS REVIEW / HIDDEN	VISIBLE TO PUBLIC
18	1	17

Search text, area, category...

24 Waterlogging & Drainage · Dwarka **OFFENSIVE LANGUAGE DETECTED (IDIOT)**

These idiot officers never clean the drain near Dwarka mor

ID 90371180 · English · 18/6/2020, 4:51:34 PM · no photo

LOG / ADMIN ACTIVITY LATEST 40

admin-login 4:51:35 PM

Figure 3. Admin console: a report held for offensive language, with Delete and Approve actions.

Screenshots show the prototype with sample data. The interactive map tab (Leaflet with OpenStreetMap) is part of the live app and is not shown here.

10 Tools and AI assistance

The prototype's source code and the drafting of this report were produced with substantial help from an AI assistant (Claude, by Anthropic), under the author's direction. The prototype also uses an AI model while running, when an API key is provided. The figures in Section 3 were checked against the source pages on 6 October 2026.

11 References

- [1] Praja Foundation (July 2020). *Report on Status of Civic Issues in Delhi*. [praja.org/praja_docs/praja_downloads/Report on Status of Civic Issues in Delhi, 2020.pdf](https://praja.org/praja_docs/praja_downloads/Report%20on%20Status%20of%20Civic%20Issues%20in%20Delhi.pdf)
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- [5] ThePrint (2 Oct 2026). *Delhi's garbage mountains are finally coming down. New ones are rising*. theprint.in/ground-reports/delhi-garbage-mountains-coming-down/3059743/
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- [7] Mishra, V. (7 Jul 2024). *Why does Delhi's Minto Bridge flood with even a few hours of rain?* Down To Earth. downtoearth.org.in
- [8] The Wire (Jul 2024). *33,000 deaths attributed each year to air pollution in 10 Indian cities, 2008 to 2019: Study* (on the *Lancet Planetary Health* study, 4 Jul 2024). thewire.in
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- [11] Schoolnet India, Amazon Future Engineers and Mu Gamma Consultants (2026). *Quest4Earth 2026 concept note*. schoolnetindia.com/hackathon/

All web pages were accessed on 6 October 2026.